

**MATH 344: Exam 2**  
**Spring —<sub>2</sub> 2026**  
**03/27/2026**  
**50 Minutes**

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**Name:** \_\_\_\_\_

Write your name on the appropriate line on the exam cover sheet. This exam contains 7 pages (including this cover page) and 6 questions. Check that you have every page of the exam. Answer the questions in the spaces provided on the question sheets. Be sure to answer every part of each question and show all your work. If you run out of room for an answer, continue on the back of the page — being sure to indicate the problem number.

| Question | Points | Score |
|----------|--------|-------|
| 1        | 10     |       |
| 2        | 10     |       |
| 3        | 10     |       |
| 4        | 10     |       |
| 5        | 10     |       |
| 6        | 10     |       |
| Total:   | 60     |       |

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1. (10 points) Let  $\mathbf{b} = \begin{pmatrix} 4 \\ 7 \\ -6 \end{pmatrix}$  and  $A$  be the matrix with LU factorization

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & -5 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$

(a) Compute  $\det A$ .

(b) Use the LU factorization above to solve the matrix-vector equation  $A\mathbf{x} = \mathbf{b}$ .

2. (10 points) Consider the invertible matrix  $A = \begin{pmatrix} 1 & -1 & -2 \\ -2 & 2 & 6 \\ 4 & -3 & -5 \end{pmatrix}$ .

(a) Explain why  $A$  *does not* have an LU decomposition without computing a PLU decomposition for  $A$ , i.e. do not use (b).

(b) Find a PLU decomposition for the matrix  $A$ .

3. (10 points) A matrix  $A$  and its reduced-row echelon form,  $R$ , are given below.

$$A = \begin{pmatrix} 4 & -1 & 5 & -4 \\ 3 & -2 & 5 & -3 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 0 \end{pmatrix}$$

(a) Determine which  $\mathbb{R}^n$  each of the spaces below are a subspace of and find their dimension.

|                | $\text{Col}(A)$ | $\text{Row}(A)$ | $\text{Null}(A)$ |
|----------------|-----------------|-----------------|------------------|
| Subspace of... |                 |                 |                  |
| Dimension      |                 |                 |                  |

(b) Find a basis for  $\text{Col}(A)$ .

(c) Find a basis for  $\text{Row}(A)$ .

(d) Find a basis for  $\text{Null}(A)$ .

4. (10 points) Complete the parts given below.

(a) Compute the following *without* explicitly computing the determinant:  $\det \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -3 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ .

(b) Find an elementary matrix that scales the first row of a  $2 \times 2$  matrix by  $\frac{1}{5}$ .

(c) If  $E = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{pmatrix}$  and  $A$  is a  $3 \times 3$  matrix, describe the matrix  $EA$ .

(d) Compute the following *without* explicitly multiplying the matrices:

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 11 & -15 & 18 & -20 \\ 8 & 5 & 16 & -9 \\ 0 & -3 & 10 & 1 \\ 3 & 17 & 0 & 4 \end{pmatrix}$$

5. (10 points) Complete the following parts:

(a) Is the set  $\left\{ \begin{pmatrix} -2 \\ 4 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -2 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -6 \\ 6 \\ 1 \\ 0 \end{pmatrix} \right\}$  linearly independent? Explain.

(b) Can a set of 12 vectors in  $\mathbb{R}^7$  be linearly independent? Explain.

(c) Find a basis for the subspace  $W = \{(x, y, z) \in \mathbb{R}^3 : 2x - y + z = 0\}$  and find its dimension.

6. (10 points) Mark the following statements True ( $T$ ) or False ( $F$ ). You *do not* need to justify your answer.

(a) \_\_\_\_\_: The columns of *any*  $6 \times 7$  matrix are linearly dependent.

(b) \_\_\_\_\_: If  $A$  is a  $3 \times 4$  matrix with rank 3, then  $A\mathbf{x} = \mathbf{b}$  *always* has a solution but the solution is *never* unique.

(c) \_\_\_\_\_: If  $\text{nullity}(A) = 0$ , i.e.  $\dim \text{null } A = \{0\}$ , then  $A\mathbf{x} = \mathbf{0}$  does not have a solution.

(d) \_\_\_\_\_: The number of independent rows and independent columns of a matrix can be different.

(e) \_\_\_\_\_: A  $12 \times 9$  matrix  $A$  can have rank at most 9.